

Network Anomaly Detection API 1.0.0 OAS 3.1

/openapi.json

Detects anomalous network traffic using XGBoost trained on NSL-KDD.

default

GET

/health

Health

Parameters

Cancel

Name	Description
x-from	x-from string (string null) (header)

Execute

Clear

Responses

Curl

curl -X 'GET' \ 'http://localhost:8000/health' \ -H 'accept: application/json' \

Request URL

http://localhost:8000/health

Server response

Code	Details
400	Error: Bad Request

Response body

{

"detail": "X-From header is required."

}

Response headers

content-length: 39
content-type: application/json
date: Fri, 19 Jul 2024 03:33:45 GMT
server: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

"string"

422

Validation Error

No links

Media type

application/json

Example Value | Schema

{

"detail": [

{

"loc": [

"string",

0

],

"msg": "string",

"type": "string",

"input": "string",

"ctx": {}

}

]

}

POST

/predict

Predict

Parameters

Try it out

Name	Description
x-from	x-from string (string null) (header)

Request body

required

application/json

Example Value | Schema

{

"urgent": 0,

"hot": 0,

"num_failed_logins": 0,

"logged_in": 1,

"num_compromised": 0,

"root_shell": 1,

"su_attempted": 1,

"num_root": 0,

"num_file_creations": 0,

"num_shells": 0,

"num_access_files": 0,

"num_outbound_cmds": 0,

"is_host_login": 1,

"is_guest_login": 1,

"count": 0,

"srv_count": 0,

"srv_error_rate": 1,

"srv_error_rate": 1,

"error_rate": 1,

"srv_error_rate": 1,

"same_srv_rate": 1,

"diff_srv_rate": 1,

"srv_diff_host_rate": 1,

"dst_host_count": 0,

"dst_host_srv_count": 0,

}

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

{

"prediction": 0,

"label": "string",

"probability": 0

}

422

Validation Error

No links

Media type

application/json

Example Value | Schema

{

"detail": [

{

"loc": [

"string",

0

],

"msg": "string",

"type": "string",

"input": "string",

"ctx": {}

}

]

}

POST

/predict/batch

Predict Batch

Parameters

Try it out

Name	Description
x-from	x-from string (string null) (header)

Request body

required

application/json

Example Value | Schema

{

"records": [

{

"duration": 0,

"protocol_type": "icmp",

"service": "string",

"flag": "string",

"src_bytes": 0,

"dst_bytes": 0,

"land": 1,

"wrong_fragment": 0,

"urgent": 0,

"hot": 0,

"num_failed_logins": 0,

"logged_in": 1,

"num_compromised": 0,

"root_shell": 1,

"su_attempted": 1,

"num_root": 0,

"num_file_creations": 0,

"num_shells": 0,

"num_access_files": 0,

"num_outbound_cmds": 0,

"is_host_login": 1,

"is_guest_login": 1,

}

]

}

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

{

"predictions": [

{

"prediction": 0,

"label": "string",

"probability": 0

}

],

"total": 0

}

422

Validation Error

No links

Media type

application/json

Example Value | Schema

{

"detail": [

{

"loc": [

"string",

0

],

"msg": "string",

"type": "string",

"input": "string",

"ctx": {}

}

]

}

Schemas

```
BatchRequest ^ object
  records ^ array-object< [1, 1000] items>
    items ^ object
      A single NSL-KDD-format network connection record.
      duration ^ number [0]
      protocol_type ^ string matches "(tcp|udp|icmp)"
      service ^ string
      flag ^ string
      src_bytes ^ number [0]
      dst_bytes ^ number [0]
      land ^ integer [0, 1]
      wrong_fragment ^ number [0]
      urgent ^ number [0]
      hot ^ number [0]
      num_failed_logins ^ number [0]
      logged_in ^ integer [0, 1]
      num_compromised ^ number [0]
      root_shell ^ integer [0, 1]
      su_attempted ^ integer [0, 1]
      num_root ^ number [0]
      num_file_creations ^ number [0]
      num_shells ^ number [0]
      num_access_files ^ number [0]
      num_outbound_cmds ^ number [0]
      is_host_login ^ integer [0, 1]
      is_guest_login ^ integer [0, 1]
      count ^ number [0]
      srv_count ^ number [0]
      error_rate ^ number [0, 1]
      srv_error_rate ^ number [0, 1]
      error_rate ^ number [0, 1]
      same_srv_rate ^ number [0, 1]
      diff_srv_rate ^ number [0, 1]
      srv_diff_host_rate ^ number [0, 1]
      dst_host_count ^ number [0]
      dst_host_srv_count ^ number [0]
      dst_host_same_srv_rate ^ number [0, 1]
      dst_host_diff_srv_rate ^ number [0, 1]
      dst_host_same_src_port_rate ^ number [0, 1]
      dst_host_srv_diff_host_rate ^ number [0, 1]
      dst_host_error_rate ^ number [0, 1]
      dst_host_srv_error_rate ^ number [0, 1]
      dst_host_error_rate ^ number [0, 1]
      dst_host_srv_error_rate ^ number [0, 1]
```

BatchResponse > object

HTTPValidationError > object

NetworkRecord > object

PredictionResponse > object

ValidationError > object